

MTE 204 – Numerical Methods

Tutorial #5 Solutions

Question 1:

Compute

$$\int_{0.5}^{2.5} \frac{1}{x^2} dx$$

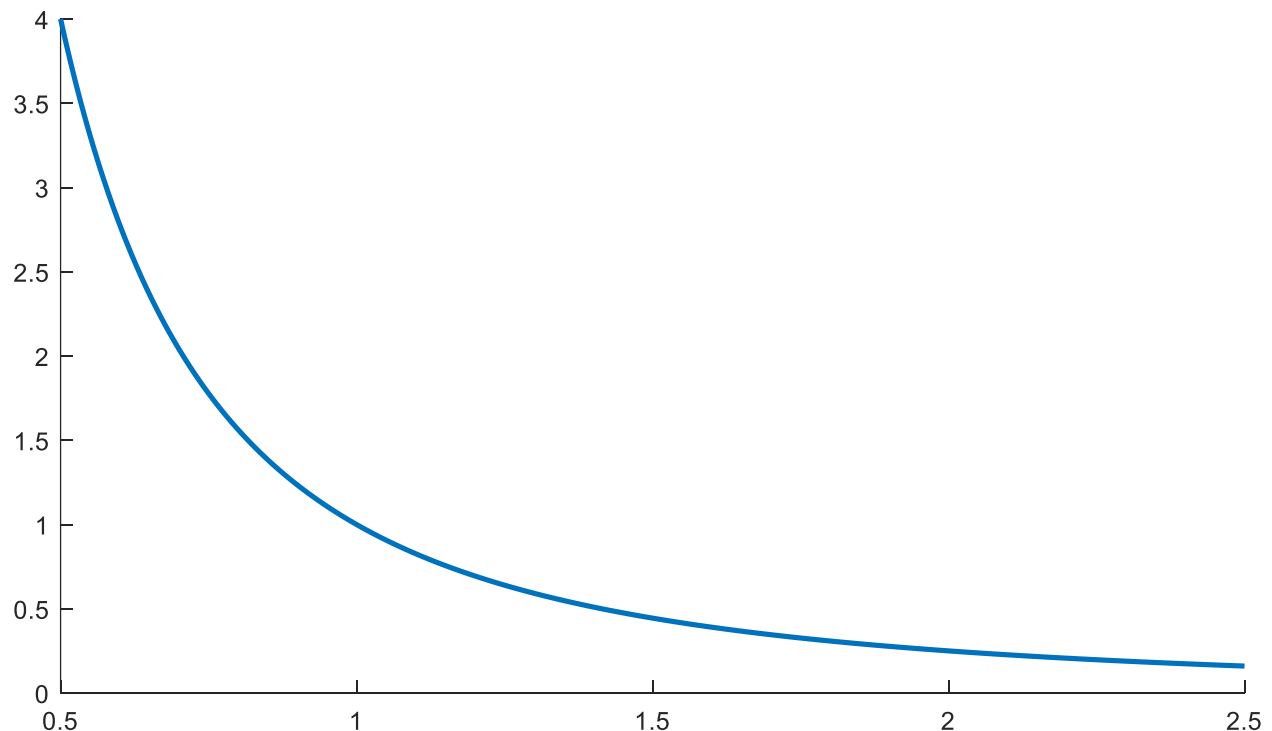
Using

- a) Trapezoid rule with $n = 1, 2, 4$, and 8 segments
- b) Romberg integration with $n = 1, 2, 4$, and 8 segments
- c) Two-point Gauss quadrature

For each method, discuss convergence behaviour, the number of function evaluations, and the error of the integral approximation.

Solution

Before we begin, let us first examine the behaviour of $\frac{1}{x^2}$ over $[0.5, 2.5]$ by plotting the function. As we can see from the figure below, this function exhibits a significant amount of non-linear behaviour.



Additionally, let us also consider the analytical solution to this problem. This will allow us to gauge the effectiveness of the three integration methods. Please note that it is not always possible to compute the analytical derivative, hence the need for numerical integration techniques.

$$\int_{0.5}^{2.5} \frac{1}{x^2} dx = -\frac{1}{x} \Big|_{0.5}^{2.5} = 1.6$$

Part A: Trapezoid Rule

Recall the general equation for integration by trapezoid rule is given by

$$I = \frac{h}{2} \left(f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n) \right)$$

Where n is the number of segments and h is the width of those segments given by the following, where a and b are the lower and upper limits of integration, respectively.

$$h = \frac{b - a}{n}$$

Let us compute the integral using a single interval ($n=1$). A single interval will lead to a step size h of 2. Further, because $n=1$, we do not need to perform the summation seen in the general equation above.

$$I_{n=1} = \frac{2}{2} \left(\frac{1}{(0.5)^2} + \frac{1}{(2.5)^2} \right) = 4.16$$

Trapezoid rule with a single segment requires two function evaluations and results in an integral that is significantly different from the analytical solution. Referring back to the plot of $\frac{1}{x^2}$, one can see that a single trapezoidal section between the two limits of integration will be a dramatic overestimation of the actual integral.

Let us now try two segments ($n=2$), which will produce a step size h of 1.

$$I_{n=2} = \frac{1}{2} \left(\frac{1}{(0.5)^2} + 2 \left(\frac{1}{(1.5)^2} \right) + \frac{1}{(2.5)^2} \right) = 2.5244$$

By adding a second segment, our error compared to the analytical solution is almost reduced by half. While only adding a single additional evaluation of the function.

Trapezoid rule is evaluated in the following table for the requested numbers of segments, where ϵ_T is the error of each numerical integration against the actual analytical function ($\epsilon_T = \left| \frac{I_{num} - I_{true}}{I_{true}} \right| \times 100\%$). Note that in many instances, one will not be able to evaluate the actual error ϵ_T as an analytic solution may not be possible. In such cases, one would need to evaluate how accurate your estimate is by performing a convergence study. Simply put, one would repeatedly double the number of segments used for integration until the difference between subsequent iterations drops below the desired tolerance.

<i>n</i> SEGMENTS	<i>I</i>	ϵ_T	NO. FUN. EVALS.
1	4.1600	160%	2
2	2.5244	57.78%	3
4	1.8872	17.95%	5
8	1.6791	4.94%	9
16	1.6204	1.28%	17

Part B: Romberg Integration

Romberg Integration (also called Romberg's method) is a technique that extends upon the trapezoid method not by improving the accuracy to which trapezoid rule can approximate an integral but rather by using multiple evaluations of trapezoid rule to produce a better estimate of the actual value.

To perform Romberg integration, we first estimate the integral of a function using the trapezoid rule twice, with n and $2n$ segments. It is very important that the number of segments is being doubled between the two iterations. Here, let us consider the case of $n = 1$ and $n = 2$, both of which have already been computed in part A ($I_{n=1} = 4.16$ and $I_{n=2} = 2.5244$). Using Romberg's Method, we can produce an estimate for the actual error using the following

$$I \cong \frac{4}{3}I_{2n} - \frac{1}{3}I_n$$

$$I \cong \frac{4}{3}(2.5244) - \frac{1}{3}(4.16) = 1.9793$$

Here we can see that we can use two approximate integrals which are clearly far from the actual value to produce a new estimate that improves upon both

This idea of successive improvement demonstrated here can be extended even further with more iterations (doubling the number of segments each time). The generalized formula for Romberg integration is given below, where j represent each successive iteration (where n is doubled) and k represents the order of the estimate, or put another way, the number of iteration used to produce an estimate

$$I_{j,k} \cong \frac{4^{k-1}I_{j+1,k-1} - I_{j,k-1}}{4^{k-1} - 1}$$

This equation is best described graphically with the use of a table. In the leftmost columns of the table below, we have the number of segments (which are doubling with each successive iteration) and the integral evaluated with the trapezoid rule. Here, the arrows denote how subsequent iterations are successively combined.

Trapezoid Rule (Initial Setup)		Romberg Integration			
Iteration j (n segments)	I_{trap}	$k = 2$ $I_{j,2}$	$k = 3$ $I_{j,3}$	$k = 4$ $I_{j,4}$	$k = 5$ $I_{j,5}$
1 (1)	4.1600	→ 1.9792	→ 1.6545	→ 1.6046	→ 1.6002
2 (2)	2.5244	→ 1.6748	→ 1.6053	→ 1.6002	→
3 (4)	1.8872	→ 1.6097	→ 1.6002	→	
4 (8)	1.6791	→ 1.6008	→		
5 (16)	1.6204	→			

Looking across the first row of the table, we can see that Romberg integration produces a more accurate estimate of the integral than any separate evaluation of the trapezoid rule. However, Romberg's method requires one to repeatedly evaluate the trapezoid rule, resulting in a large number of function evaluations. That said, to perform a convergence study (as mentioned in part A), one will likely have already performed these evaluations. Thus, Romberg's method lets us use these evaluations more effectively.

Part C: Gauss Quadrature

Gauss quadrature, also known as Gauss-Legendre integration, is similar in concept to the trapezoid rule. However, instead of evaluating the function at the end of each segment, we use special points inside each segment known as gauss points. This technique can help accommodate for non-linearities that may occur across the segment.

Gauss-Legendre integration requires some preliminary manipulation of the function prior to performing the actual integration. Specifically, Gauss-Legendre integration requires the function to be evaluated over $[-1, 1]$, not the limits of integration that may be specified in the problem. To accommodate this, we map the limits of integration given on the problem to $[-1, 1]$ using the following transformation

$$x = \frac{(b + a) + (b - a)x_d}{2}$$

Where x_d is the variable corresponding to the Gauss-Legendre limits of $[-1, 1]$. For this problem, we have the following mapping function.

$$x = \frac{(2.5 + 0.5) + (2.5 - 0.5)x_d}{2}$$

$$x = 1.5 + x_d$$

We also need to transform the differential of x , which can be accomplished with the following formula

$$dx = \frac{b - a}{2} dx_d$$

$$dx = \frac{2.5 - 0.5}{2} dx_d = dx_d$$

We can insert this mapping function into our original function to transform the limits of integration of $[0.5, 2.5]$ to $[-1, 1]$

$$\int_{0.5}^{2.5} \frac{1}{x^2} dx = \int_{-1}^1 \frac{1}{(1.5 + x_d)^2} dx_d$$

The equation for 2-point Gauss-Legendre Integration is as follows, where the function is evaluated in the transformed space of $[-1, 1]$. Compared to the trapezoid rule, a simple summation is all that is required due to our transformation of the limits of integration.

$$I \cong f\left(-\frac{1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right)$$

For this problem in specific, we get the following

$$I \cong \frac{1}{\left(1.5 + \left(-\frac{1}{\sqrt{3}}\right)\right)^2} + \frac{1}{\left(1.5 + \left(\frac{1}{\sqrt{3}}\right)\right)^2}$$

$$I \cong 1.4064$$

With only a single segment, the integral is approximated as 1.4064. Furthermore, this result was achieved with only two evaluations of the function being integrated. This is a dramatic improvement over the previous two methods discussed in this tutorial.

While this question only specified two-point Gauss-Legendre integration, this technique can be extended to many more points for a more accurate estimate of the integral. Increasing the number of points will change the location of the Gauss points and the coefficients used when summing function evaluations together. The derivation of these points and coefficients is beyond the scope of this course, but tables of values can be found in the course text (Table 22.1) or online sources like Wikipedia. An example of how increasing the number of Gauss-Legendre points can improve the approximation of the integral used in this problem is given below. The improvement with each additional point is remarkable.

NUMBER OF GAUSS- LEGENDRE POINTS	COMPUTED INTEGRAL
2	1.4064
3	1.5582
4	1.5919

From a practical perspective, Gauss-Legendre is often performed using multiple segments like the trapezoid rule. However, Gauss-Legendre integration generally requires far fewer segments. Like the trapezoid rule, a convergence study is required when the analytical function is not known. As an extension, Romberg's method can be extended to use Gauss-Legendre integration to extrapolate an even more accurate estimate of the integral in such cases.